

1. What is URL? (2 marks)

URL (Uniform Resource Locator) is the address used to access resources on the internet.
Example: <https://www.google.com>

2. Define HTTP cache. (2 marks)

HTTP cache is a mechanism that stores copies of web resources to reduce server load, improve speed, and reduce latency when the same resource is requested again.

3. What is a cookie? (2 marks)

A cookie is a small piece of data stored on the user's browser by websites to remember information such as login status, preferences, or activity.

4. What is a confusion matrix? (2 marks)

A confusion matrix is a table used to evaluate the performance of a classification model by showing the actual vs predicted results in

categories like True Positive, False Positive, True Negative, and False Negative.

5. Name any 4 web designing tools. (2 marks)

1.
Adobe Dreamweaver
 2.
Figma
 3.
WordPress
 4.
Visual Studio Code
-

6. What is sentiment analysis? (2 marks)

Sentiment analysis is a technique in Natural Language Processing (NLP) used to determine whether a text expresses a **positive**, **negative**, or **neutral** sentiment.

7. Define cookies. (2 marks)

Cookies are **text files** stored in the browser to store user-related data, like preferences or session information, which helps personalize the user experience.

8. What is scripting language? (2 marks)

A **scripting language** is a programming language used to write scripts that automate tasks on websites.
Examples: JavaScript, PHP, Python.

9. Define: (2 marks)

- a) **Internet**: A global network of computers that enables sharing of information and communication.
 - b) **WWW (World Wide Web)**: A collection of web pages accessible via the Internet using web browsers.
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10. What is protocol? Name any two. (2 marks)

A **protocol** is a set of rules that defines how data is transmitted over a network.

Examples:

- HTTP (Hypertext Transfer Protocol)
 - FTP (File Transfer Protocol)
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11. What is domain name? Name any two. (2 marks)

A **domain name** is the human-readable address of a website, used instead of IP addresses.

Examples:

- google.com
- wikipedia.org

12. Name any 4 web designing tools. (*Repeated*)

1.
Adobe XD
2.
Sketch
3.
Canva
4.
Webflow

13. Explain HTTP request. Mention any four methods.

HTTP (**H**yper**T**ext **T**ransfer **P**rotocol) is the protocol used by web browsers and servers to communicate. It allows clients (like web browsers) to request resources from a server (like HTML files, images, or data). An HTTP request contains a method, URL, headers, and sometimes a body.

Four Common HTTP Request Methods:

1.

GET – Retrieves data from the server (e.g., a webpage).

2.

POST – Sends data to the server, often used in form submission.

3.

PUT – Updates existing data on the server.

4.

DELETE – Deletes data from the server.

14. What is client-side scripting and server-side scripting? Explain the differences.

Client-side scripting is code that runs on the user's browser. Examples: JavaScript, HTML, CSS.

Server-side scripting runs on the server and generates dynamic content. Examples: PHP, Node.js, Python.

Differences:

Aspect	Client-Side Scripting	Server-Side Scripting
Execution	On the user's browser	On the web server
Language	JavaScript, HTML	PHP, Node.js, Python
Speed	Faster as no server contact is needed	Slower due to server processing
Security	Less secure	More secure
Example	Form validation	Login authentication

15. Explain Web 3.0.

Web 3.0 is the third generation of web evolution, focusing on a decentralized, intelligent, and more user-friendly web experience.

Key Features:

- **Decentralization:** Data is stored in distributed networks like blockchain.
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Semantic Web: Machines understand data contextually.

- **AI and ML:** Artificial intelligence helps make smarter search and predictions.
- **Interoperability:** Applications and services can communicate across platforms.
- **Ownership and Privacy:** Users have control over their data.

Example: Using blockchain-based apps for transactions or data sharing without central authorities.

16. What is a list in HTML? Explain different types of lists with example.

In HTML, a list is used to group related items. There are **three types**:

1.
Ordered List () – Items are numbered.

html

Copy code

```
<ol> <li>HTML</li> <li>CSS</li> <li>JavaScript</li> </ol>
```

2.

Unordered List () – Items are bulleted.

html

Copy code

```
<ul> <li>HTML</li> <li>CSS</li> <li>JavaScript</li> </ul>
```

3.

Definition List (<dl>) – Used for terms and their definitions.

html

Copy code

```
<dl> <dt>HTML</dt> <dd>Hyper Text Markup Language</dd> </dl>
```

17. Explain the difference between PHP and Node.js.

Feature	PHP	Node.js
Language	Server-side scripting language	JavaScript runtime environment
Execution	Synchronous (by default)	Asynchronous (non-blocking I/O)
Hosting	Widely supported on shared hosting	Needs server or cloud platforms
Performance	Slower due to blocking I/O	Faster due to event-driven architecture
Use Case	Traditional websites, CMS (like WordPress)	Real-time apps (like chat), APIs

Conclusion: PHP is simple and best for traditional web apps; Node.js is ideal for scalable and real-time applications.

18. What is NoSQL? Explain the difference between NoSQL and SQL.

NoSQL (Not Only SQL) is a type of database that stores data in non-tabular formats like key-value, document, graph, or wide-column stores. It is designed for large-scale, distributed data systems.

Differences:

Feature	SQL (Relational DB)	NoSQL (Non-Relational DB)
Structure	Tables with rows and columns	JSON, key-value, graphs, etc.
Schema	Fixed schema	Dynamic schema
Scalability	Vertical scaling	Horizontal scaling
Best For	Structured data	Unstructured or semi-structured data
Examples	MySQL, PostgreSQL	MongoDB, Cassandra, Redis

19. What is HTML? Explain any 4 tags with example.

HTML (HyperText Markup Language) is the standard language used to create and design webpages. It structures web content using elements represented by tags.

Four Common HTML Tags:

1.

`<html>` – Root element of the page.

```
html
```

Copy code

```
<html> ... </html>
```

2.

`<head>` – Contains metadata and links.

html

Copy code

```
<head><title>Page</title></head>
```

3.

<body> – Main content of the page.

html

Copy code

```
<body>Welcome!</body>
```

4.

<a> – Anchor tag for hyperlinks.

html

Copy code

```
<a href="https://example.com">Click here</a>
```

20. Explain HTTP request method.

An **HTTP request method** defines the action to be performed on a given resource.

Common Methods:

1.

GET – Retrieves data from the server.

2.

POST – Sends data to be processed.

3.

PUT – Updates a resource.

4.

DELETE – Removes a resource.

Each method allows interaction between clients (like browsers) and servers to exchange data over the web.

21. What is CSS? Explain different types of CSS.

CSS (Cascading Style Sheets) is used to style HTML elements (fonts, colors, layout).

Types of CSS:

1.

Inline CSS – Style in the HTML tag itself.

html

Copy code

```
<p style="color:red;">Text</p>
```

2.

Internal CSS – Defined within <style> tag inside the HTML <head>.

html

Copy code

```
<style> p { color: blue; } </style>
```

3.

External CSS – Written in a separate .css file and linked.

html

Copy code


```
<link rel="stylesheet" href="style.css">
```

22. Differentiate client and server side scripting.

(Already answered above in Q8)

23. Explain the applications of Internet.

Internet applications are the various services and platforms we use online.

1.

Communication: Email, chat, video conferencing.

2.

Information access: Search engines, online libraries.

3.

Social networking: Facebook, Instagram.

4.

E-commerce: Online shopping (Amazon, Flipkart).

5.

E-learning: Online courses, certifications.

6.

Banking & Finance: Internet banking, UPI.

24. What is online certification? Explain.

Online certification is a process of earning a certificate by completing a course via the internet. These are offered by platforms like Coursera, Udemy, and edX.

Benefits:

-

Flexibility in learning.

-

Skill development and career growth.

- Globally recognized certifications.

Example: Google IT Support Certificate, AWS Cloud Practitioner Certificate.